

Total No. of Questions : 8]

SEAT No. :

PE-4294

[Total No. of Pages : 2

[6582]-67

S.E. (Artificial Intelligence and Machine Learning)

COMPUTER GRAPHICS

(2019 Pattern) (Semester - IV) (218555)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) **Answers : Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.**
- 2) **Neat diagrams must be drawn wherever necessary.**
- 3) **Figures to the right side indicate full marks.**
- 4) **Assume suitable data if necessary.**

Q1) a) Explain the basic transformation techniques in 3D Graphics : [9]

- i) Rotation
- ii) Translation
- iii) Reflection about XY Plane

b) Let ABCD be the rectangle window with A (150, 150), B (150, 200), C (200, 200) and D (200, 150). Use Cohen Hodgeman polygon clipping algorithm to clip the convex polygon PQR with P (170,250), Q (180, 100), R (250, 180) and find the final coordinates of the clipped polygon. **[9]**

OR

Q2) a) Let ABCD be the rectangle window with A (20, 20), B (60, 20), C (60, 40) and D (20,40). Find the region code for endpoints and use Cohen Sutherland algorithm to clip the lines P1-P2 with P1 (10, 25) and P2 (70, 35) and Q1-Q2 with Q1 (30, 10) and Q2(50, 70) Show Graphic Representation of Original and Clipped Line. [9]

b) What is projection? Explain in detail with diagram parallel and perspective projection. **[9]**

Q3) a) What is a segment? Why do we need segments? Explain the complete process of i) Segment Creation, ii) Segment Deletion and iii) Segment Renaming. [9]

- b) Explain in detail with Diagram **[8]**
- i) RGB Color Model
 - ii) CMY Color Model
 - iii) HSV Color Model
 - iv) Color Gamut

P.T.O.

OR

- Q4)** a) Compare the following illumination models with diagrams i) Ambient Light, ii) Diffuse Light and iii) Specular reflection [9]
- b) Define Shading. Explain with help of diagrams Phong Shading algorithm in detail. [8]

- Q5)** a) Explain in detail with diagram Bezier curve generation using midpoint subdivision method. [9]
- b) Write short note on : [9]
- i) Fractals and topological dimensions.
 - ii) Koch Curve
 - iii) Hilbert Curve

OR

- Q6)** a) Write Short note on : [9]
- i) Methods of controlling animation?
 - ii) Types of animation languages.
- b) What is interpolation? Explain in detail with diagram the process of curves Approximation. [9]

- Q7)** a) Explain the behavior modeling in Virtual Reality. [6]
- b) Explain haptic feedback in Virtual Reality system. [6]
- c) Differentiate HMD and CAVE in Virtual Reality. [5]

OR

- Q8)** a) Explain the physical modeling in Virtual Reality. [6]
- b) What is geometric modeling in a Virtual Reality? [6]
- c) Explain gesture interfaces in Virtual Reality. [5]

